Colour Contrast Check (NFR-08)

Prepared ahead of Task 3.16 · v1.0 · Owner: Syed Sarim Jawad Zaidi · 21 Oct 2026 · Checked: style guide + all 63 hi-fi screens on dev (PR #53) + Talha's app colours (main, 55ab1d6)

Rule used: NFR-08 asks for WCAG 2.1 Level AA contrast on the three safety-critical Dependent screens (Panic, Wellness check, Medicine) in Large text / High-contrast mode, as a best-practice benchmark. AA means a contrast ratio of at least 4.5 : 1 for normal text and 3 : 1 for large text (24 px and up, or about 19 px and up in bold). This check applies the same rule to every screen, so nothing needs fixing later.

How: a script read every text element in the 63 mockups, found the colour behind it and calculated the ratio with the WCAG formula. Status-bar text (time, battery) is drawn by Android, not by our app, and is ignored.

1. Result in one line

The brand colours pass. Teal #0E7490 with white is 5.36 : 1, dark text on white is 17.85 : 1. 24 text spots on 4 colour pairs are below AA, including the small line under the PANIC button on three safety-critical screens. Two token changes fix all of them.

2. Pairs below AA in the mockups

Text colour on background	Ratio	Needs	Where (examples)	Spots	Fix
#FEE2E2 (light red) on #DC2626 (Panic red)	3.95	4.5	Line under PANIC: 20 Home, 21 Panic, 29 Home; 07 Home panic banner	4	Danger token → #B91C1C (5.30); or white text
#DC2626 (red) on #FEE2E2 (red tint)	3.95	4.5	02 Log in error banner	2	Danger token → #B91C1C (5.30)
#64748B (grey) on #E2E8F0 (unselected tab)	3.86	4.5	Segmented tabs: 03, 08, 11c, 13, 15, 15d, 16, 17, 17d, 18, 18b, 26	14	Muted token → #475569 (6.15)
#64748B (grey) on tinted cards (red, amber, green tints)	3.83–4.27	4.5	21b time, 11c timeline, 17b and 25c urgent-voice times	4	Muted token → #475569 (6.20–6.81)

Safety-critical screens: 20, 21 and 29 fail only on the small line under PANIC ("Press for help", "to send help"). Everything else on 20–24 passes (white on teal 5.36, white on dark teal 7.27, dark text 17+).

3. Pairs that pass but are close to the limit

Pair	Ratio	Note
#B45309 amber on #FEF3C7 amber tint	4.51	Passes; the darker muted token does not affect it. Keep amber for titles only, not for small text.
#64748B grey on #F8FAFC page background	4.55	Passes; becomes 7.24 with the new muted token.
#15803D green on #DCFCE7 green tint	4.57	Passes; keep for short labels (Verified, Inside).
#64748B grey on #ECFEFF teal tint	4.57	Passes; improves with the new muted token.

4. The two token changes (style guide)

Token	Now	New	Why
danger (Panic red)	#DC2626	#B91C1C	White on it goes from 4.83 to 6.47; the light-red line under PANIC from 3.95 to 5.30; red text on the red tint from 3.95 to 5.30. Still clearly red.
muted (secondary text)	#64748B	#475569	Grey text passes on every background we use (6.15 or more), including unselected tabs and tinted cards.

All other tokens stay the same. The mockup PNGs keep the old two colours until each set gets its next version; the style guide is the reference for the app.

5. App build (Talha's repo) — same check, for information

Where in the app	Colours	Ratio	Result
Panic button subtitle (Colors.white70 on Colors.red.shade700)	#F2C1C1 on #D32F2F	3.12	Below AA on a safety-critical screen. Use white, or the new danger #B91C1C.
Location banner text and outlines (Colors.teal)	#009688 on white	3.67	Below AA for text. The style guide teal #0E7490 gives 5.36.
Theme seed colour (buttons, app bar)	white on #2E7D32 (green)	5.13	Passes, but it is green, not the style guide teal.
Panic button label (white on Colors.red.shade700)	#FFFFFF on #D32F2F	4.98	Passes.

These are non-blocking notes for Talha's UI polish; they are recorded in the Full Task List (rows 1.9 and 3.10).

6. Still to check on a real device (Task 3.16)

- Large text / High-contrast mode switched ON on screens 20–24: ratios, text not cut off, buttons still reachable.
- NFR-07: Panic, I am OK and My medicine each completed within 2 taps from the Dependent home.
- Urdu text in the same places (longer words, right-to-left).